AN INTERACTIVE DIGITAL FILTER WORKBENCH

Make the connection.

Move a pole. Place a zero. See what changes in frequency and time.

Export filter | Import JSON

Sampling rate 10 Hz | Nyquist 5 Hz | Experiment: Custom filter | Reset experiment

01 / BUILD YOUR FILTER

LIVE

H(z) | Poles & zeros | Design in Hz

Start with a frequency

Choose a response shape, then explore the filter that creates it.

Design method: Standard filter

Family: IIR · biquad

Response: Low-pass

Cutoff frequency (Hz): 1.4

Q · sharpness: 0.70710678118655

Second-order biquad. Low/high-pass cutoff is −3 dB when Q = 0.7071. For band-pass/notch, Q controls bandwidth: higher Q gives a narrower response.

Design filter ↗

TRY THIS

These poles create a recursive response. Save a comparison, then change Q: sharper frequency selectivity generally comes with longer ringing in time.

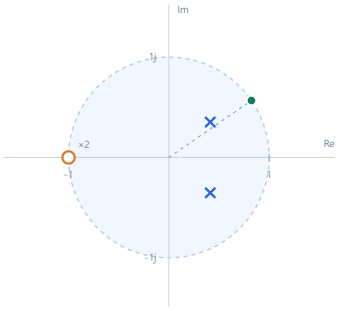
02 / EXPLORE THE RESPONSE

IIR low-pass · 1.4 Hz

IIR recurrence · order 2 | Stable

z-plane

UNIT CIRCLE



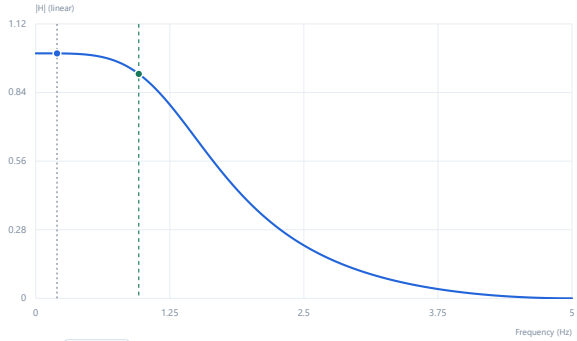
ω = 0.192π rad/sample

Zero | Pole | Frequency

2 effective zero(s) · 2 effective pole(s), including origin markers. Positions recovered from coefficients. Drag or use the numeric editor.

Magnitude response

dB | Linear



Frequency axis: Hz | + Save comparison (0/5)

Hover to compare · Click to set reference.

▸ Axes — Full frequency range · Auto magnitude

Inspect frequency: 0.960521 Hz — 0.9161 × — −58.98° · wrapped phase

Compare attenuation

Compare at: 0.19828131 Hz

Reference: 0.9605218 Hz

Compare at: 0.1982813 Hz

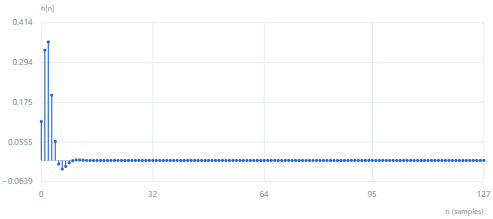
Curve	Reference · dB	Compare at · dB	Δ gain · dB	Amplitude ratio
Live · IIR low-pass · 1.4 Hz f, 10 Hz · 0.9605218 → 0.1982813 Hz	−0.76072	−0.0013403	0.75938 0.75938 dB less attenuation	1.0914×

Negative Δ gain means more attenuation relative to the reference. −20 dB = 0.1× amplitude. The two gain columns are absolute gains relative to the input.

Each curve is measured at the same frequencies in Hz, within its own Nyquist range.

Impulse response

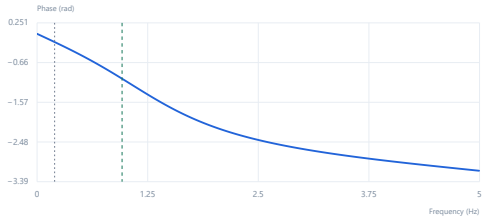
Samples: 128



First 128 samples, starting from zero initial conditions. An IIR tail may extend beyond this view.

Phase response

UNWRAPPED · RADIANS



Gaps mark zero magnitude or a singular response, where phase is undefined.

The filter behind the plots

Copy coefficients

$$H(z) = \frac{0.117351 + 0.234702z^{-1} + 0.117351z^{-2}}{1 - 0.825232z^{-1} + 0.294637z^{-2}}$$

DIFFERENCE EQUATION

$y[n] = 0.11735 \cdot x[n] + 0.2347 \cdot x[n-1] + 0.11735 \cdot x[n-2] + 0.82523 \cdot y[n-1] - 0.29464 \cdot y[n-2]$

WHAT THIS MEANS

IIR feeds past outputs back into the recurrence. Its impulse usually continues indefinitely. For a causal recurrence, all denominator poles must lie strictly inside the unit circle for asymptotic stability.

▾ Full precision coefficients

```
b = [0.11735103672460925, 0.2347020734492185, 0.11735103672460925]
a = [1, -0.8252323806894792, 0.2946365275879162]
fs = 10 Hz
```


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